Lecture 7

- Least Squares Solution
- Optimization of general quadratic functions

Consider a linear system with m eq. and n unknowns

$$Ax = b \qquad (1)$$

$A \in \mathbb{M}_{m\times n}$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$

Suppose it does not have a solution : $b \notin \operatorname{im} A$

$\Updownarrow$

$\operatorname{rank}[A|b] > \operatorname{rank} A$

usually for tall matrices $\boxed{A}$ $m > n$

In applications, we still need a "solution".

<u>Idea:</u> solving (1) is matching Ax with b.
if we can't do this exactly, let's do this approx.

Let $\underbrace{r}_{\text{residual}} = Ax - b$. Let's find x such that $\|r\|$ is as small as possible.

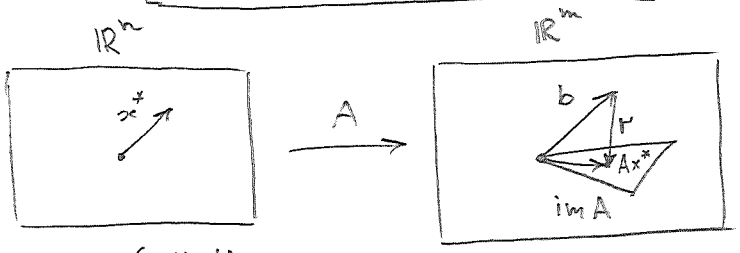
<u>Def</u> The least squares solution of $Ax = b$ is

$$\boxed{x^* = \arg\min_{x\in\mathbb{R}^n} \|Ax - b\|}$$

Euclidean norm

$x \in \mathbb{R}^n$

$Ax \in \operatorname{im} A$



(exact)

<u>Remark</u>: An ordinary solution is also a least squares solution.

Let's try to solve $\|Ax-b\|^2 \to \min$

$$\|Ax-b\|^2 = \langle Ax-b, Ax-b\rangle = \langle Ax, Ax\rangle - 2\langle Ax, b\rangle + \|b\|^2$$

$$= (Ax)^T Ax - 2(Ax)^T b + \|b\|^2$$

$$= x^T \underbrace{A^T A}_{K} x - 2x^T \underbrace{A^T b}_{f} + \underbrace{\|b\|^2}_{c} \leftarrow \text{quadratic function of } (x_1 \dots x_n)$$

$$\|Ax-b\|^2 = x^T K x - 2x^T f + c \to \min$$

- $K = A^T A$, the Gram matrix associated with $\underbrace{a_1, \dots, a_n}_{\text{columns of } A}$, $K \geq 0$
- $f = A^T b$
- $c = \|b\|^2$

Minimization of Quadratic Functions

$x \in \mathbb{R}^n$, $p(x) = x^T K x - 2x^T f + c \to \min$ (2) and $K^T = K$

<u>Remark</u> (2) is a general q.f. (any can be written in this form)

Indeed, $p(x) = \underbrace{\sum_{i,j=1}^n a_{ij} x_i x_j}_{} + \underbrace{\sum_{i=1}^n b_i x_i}_{} + c$

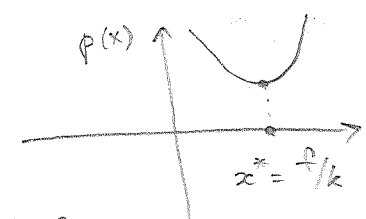
$\mathbb{R} \ni x^T A x = \qquad x^T b = -2x^T\left(\frac{b}{2}\right)$

$= \frac{1}{2}\left(x^T A x + (x^T A x)^T\right) = x^T\left(\frac{A + A^T}{2}\right)x$ ← symmetric

<u>Simple case : $n = 1$</u>

$p(x) = kx^2 - 2xf + c \to \min$

If $k > 0 \Rightarrow x^* = \frac{f}{k}$



<u>Th 1</u> If $K > 0$, then $\exists!$ global minimizer, which is the solution of

$$\boxed{Kx^* = f \iff x^* = K^{-1} f}$$

<u>Proof</u>: We need to "complete the square"

$$p(x) = x^T K x - 2x^T \underbrace{f}_{Kx^*} + c = x^T K x - 2\underbrace{x^T K x^*}_{= x^{*T} K x} + c$$

$$= x^T K x - x^T K x^* - x^{*T} K x + c =$$

$$= \underbrace{(x-x^*)^T K (x-x^*)} - \underbrace{(x^*)^T K x^* + c}_{\text{does not depend on } x}$$

$$= q_K(x-x^*) - \xi(x^*, K, c) \to \min$$

Since $K > 0$, $q_K(z) \geq 0$, $q_K(z) = 0 \iff z = 0$

$\Rightarrow$ x^* is the global minimizer of $p(x)$ $\square$

<u>Example</u> $p(x) = 4x_1^2 - 2x_1x_2 + 3x_2^2 + 4x_1 - 2x_2 + 1 \to \min$

1. $K = \begin{bmatrix} 4 & -1 \\ -1 & 3 \end{bmatrix}$ 2. $K \overset{?}{>} 0$ yes 3. Solve $Kx = f$, $f = \begin{bmatrix} -2 \\ +1 \end{bmatrix}$

<u>Q</u>: What if K is not positive def ?

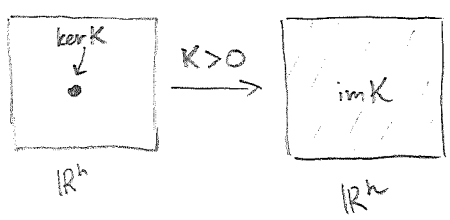
<u>Th 2</u> There are 3 cases

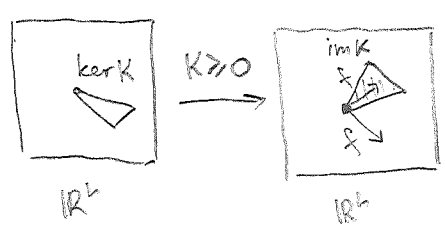
(1) $K > 0 \overset{Th1}{\Longrightarrow} \exists!$ global minimizer $x^* = K^{-1}f$

(2) $K \geq 0$ and $f \in \operatorname{im} K \Rightarrow$ a global minimizer is a solution of $Kx^* = f$ (it is <u>not</u> unique)
if $z \in \ker K \Rightarrow p(x^*) = p(x^* + z)$

(3) Otherwise, there is no global minimum.

never occur in least squares





Let's apply this result to the least squares problem

(1) $K = A^T A \geq 0$

$K > 0 \iff a_1 \dots a_n$ lin ind $\iff$ rank $A = n$ (full rank), $\ker A = \{0\}$

<u>Th 3</u> If $\ker A = \{0\}$, then the least squares sol. of $Ax = b$ is the solution of

$$\boxed{A^T A x^* = A^T b} \iff \boxed{x^* = (A^T A)^{-1} A^T b}$$

normal equations

<u>Remark</u> This is a <u>new</u> formula for the exact solution of $Ax = b$ when $b \in \operatorname{im} A$.

- $m = n \Rightarrow x^* = A^{-1} b$
- $m \neq n \Rightarrow x^* = (A^T A)^{-1} A^T b$

<u>Remark</u> Let's compute the least square error (LSE)

$$\|Ax^* - b\|^2 = \langle Ax^* - b, Ax^* - b \rangle = (Ax^*)^T A x^* - 2(Ax^*)^T b + \|b\|^2$$

$$= (x^*)^T A^T A x^* - 2(x^*)^T A^T b + \|b\|^2$$

$$= b^T A \underbrace{(A^T A)^{-1} A^T A}_{I} (A^T A)^{-1} A^T b - 2 b^T A (A^T A)^{-1} A^T b + \|b\|^2$$

$$= \|b\|^2 - b^T A (A^T A)^{-1} A^T b.$$

$$LSE = \sqrt{\|b\|^2 - b^T A (A^T A)^{-1} A^T b}$$

(2) $\ker A \neq \{0\} \iff a_1 \dots a_n$ are lin. dep.

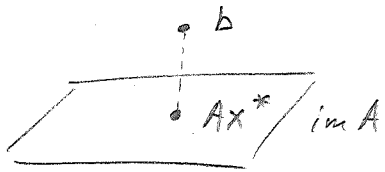
In this case, $K = \underbrace{A^T A}_{\text{singular}} \geq 0$ (and $K \not> 0$)

<u>Q</u>: Can we have $f \notin \operatorname{im} K$?

$\iff$ normal eq. don't have solution

<u>A</u>: No! since the least square sol. alway exist!

(may be not unique)
minimizer.



In this case, the least squares solution is not unique and there is no explicit formula for $x^* = \dots$ x^* is a sol of $A^T A x^* = A^T b$.

To get a "closed-form solution", in practice, regularization is often used:

$$\|Ax - b\|^2 + \lambda \|x\|^2 \longrightarrow \min$$

$\uparrow$ $\lambda > 0$, small parameter

This optimization problem is equivalent to

$$\|\tilde{A}x - \tilde{b}\|^2 \longrightarrow \min$$

$$\tilde{A} = \boxed{\begin{matrix} A \\ \sqrt{\lambda}\, I_n \end{matrix}} \in \mathbb{M}_{(m+n)\times n}, \quad \tilde{b} = \begin{matrix} \\ n\{ \end{matrix}\boxed{\begin{matrix} b \\ 0 \end{matrix}} \in \mathbb{R}^{m+n}$$

$\uparrow$ full rank ($\ker \tilde{A} = \{0\}$)

$$\Rightarrow \boxed{\tilde{x}^* = (A^T A + \lambda I_n)^{-1} A^T b}$$